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SYMBIOSIS INSTITUTE OF COMPUTER STUDIES AND RESEARCH (SICSR)
SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY), PUNE

AETHER ACADEMY

Advanced Academic Administration & Student Management System

Project Deliverable:	Web UI and Content Management Academic Syllabus
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Academic Cycle:	2026
GitHub Repository:	https://github.com/bLaZeBeCoding/Aether-Academy-LMS

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1. Executive Summary & Introduction

The continuous modernization of higher education demands a decisive transition from fragmented, error-prone manual administration toward unified, real-time web applications. Academic institutions process vast amounts of mission-critical data daily, spanning applicant admissions, curriculum planning, faculty workload distribution, and student attendance tracking. Historically, institutions have relied on decentralized spreadsheets or paper ledgers, resulting in severe data desynchronization, administrative fatigue, and a critical lack of operational transparency for learners.

Aether Academy was conceptualized and engineered to resolve these operational bottlenecks. Developed as an enterprise-grade Student Management System (LMS), the platform deploys a highly responsive "Night Owl" dark-mode user experience combined with a decoupled, procedural PHP 8 backend and a normalized MySQL relational schema. The platform establishes strict role-based data segregation between administrative staff and enrolled students, ensuring administrative superiority while empowering learners with immediate visibility into their academic trajectories.

📌 **EXECUTIVE ARCHITECTURAL TARGET**

Aether Academy eliminates paper workflows by unifying Admissions, Course Cataloging, Faculty Structuring, and Dynamic Attendance Analytics within a single secure, responsive ecosystem.

2. Problem Statement & Institutional Rationale

Traditional collegiate record-keeping models face acute operational vulnerabilities that impede organizational efficiency and student outcomes. Specifically, three core structural deficits were identified during baseline analysis:

- **Severe Data Fragmentation & Redundancy:** When admissions, student rosters, and course enrollments are tracked across isolated files, human entry errors compound. Updating a student's profile in one document fails to propagate to faculty attendance sheets, creating orphaned records.
- **Attendance Opacity & Defaulter Risk:** In legacy workflows, attendance percentages are calculated manually at the end of the semester. Students frequently discover they have fallen below the mandatory 75% attendance threshold only when they are formally debarred from examinations.
- **Authentication & Session Insecurity:** Many institutional internal portals store passwords in plaintext or employ outdated cryptographic hashes, lacking client-side idle timeouts. Shared laboratory workstations are vulnerable to unauthorized session reuse.

3. Objectives & Scope of the Project

3.1 Primary Technical Objectives

- Architect and deploy a role-based access control (RBAC) portal isolating Admin and Student operations.
- Eliminate intrusive HTTP page refreshes during admissions using asynchronous jQuery AJAX form submission.
- Implement enterprise-grade password security using BCrypt with automatic legacy migration.
- Design a normalized MySQL schema preventing duplicate attendance records using compound UNIQUE constraints.
- Construct an automated client-side idle session termination protocol (90-second timeout with animated UI toasts).

3.2 Project Scope & Operational Boundaries

The project encompasses full-stack architectural engineering. The frontend is built on semantic HTML5, CSS Grid, and Bootstrap 5. The backend utilizes procedural PHP 8 with prepared MySQLi transactions. The operational scope strictly covers Admissions, User Authentication, Teacher Profiling,

Course Structuring, Student Enrollment, and Attendance Analytics. Financial tuition collection, automated SMS/email gateways, and grading systems are designated for Future Scope.

> **SECURITY COMPLIANCE HIGHLIGHT**
>
> Client-side session monitoring actively terminates idle logins after 90 seconds, defending shared campus workstations against session hijacking.

4. Technologies Used (Full-Stack Architecture)

Aether Academy is built upon the time-tested LAMP (Linux, Apache, MySQL, PHP) architectural pattern, enhanced with modern client-side libraries to achieve high responsiveness and visual accessibility.

Technology	Layer / Role	Architectural Functionality
PHP 8	Server-Side Backend	Handles core routing, role authorization, session cryptography & MySQLi queries.
MySQL (InnoDB)	Relational Database	Stores normalized academic entities with cascading foreign keys and transactions.
HTML5 & CSS3	Semantic Presentation	Bespoke 'Night Owl' Glassmorphism theme, CSS variables, and high contrast.
Bootstrap 5.3	Responsive Grid	Mobile-first grid layout, modal systems, form components, and utility styling.
JavaScript (ES6)	Client Scripting	DOM event monitoring, synchronous FOUC prevention, and 90s idle timer.
jQuery 3.7	DOM & AJAX Framework	Asynchronous admission submission and dynamic back-button DOM injection.
DataTables 1.13	Data Presentation	Instant client-side searching, column sorting, and pagination for large rosters.
Apache 2.4	Local Web Server	HTTP request processing provided via XAMPP development environment.

5. Implementation & System Architecture

5.1 Database Entity-Relationship (ER) Architecture

The database ('collegepr') was engineered with strict normalization principles. Referential integrity is strictly maintained via foreign key constraints with ON DELETE CASCADE rules, ensuring parent record deletions automatically purge child relations.

Table Name	Primary Key	Attributes & Foreign Keys	Relational Function
user	id	username, password (BCrypt), email, phone, usertype	Central authentication node.
teacher	id	name, description, image (path)	Faculty member profiles.
course	id	name, teacher_id (FK -> teacher.id)	Curriculum linked to instructor.
enrollment	id	student_id (FK), course_id (FK)	Many-to-Many junction table.
attendance	id	student_id, course_id, attendance_date, status	Daily Present/Absent status.

5.2 Security Implementations & Session Protection

Security was prioritized across every layer of the software stack to defend academic assets against common vulnerabilities:

- **BCrypt Password Hashing:** Passwords are cryptographically salted and hashed using PHP's native `password_hash()` with cost-factor 10. A backward-compatible auto-upgrade protocol automatically upgrades legacy plaintext passwords to secure hashes upon successful login.
- **SQL Injection Defense:** All user-supplied parameters across authentication, student updates, and admission handling are escaped via `mysqli_real_escape_string()` prior to database query execution.
- **Asynchronous Idle Session Timeout:** An event listener in `admin_css.php` and `student_css.php` monitors user activity. Inactivity exceeding 90 seconds triggers an automatic redirect to `logout.php?reason=timeout`, routing to a custom 7-second animated CSS progress-slider toast on the login screen.

5.3 Multi-Layered Intellectual Property Watermarking

To defend the architecture against uncredited duplication, an immutable 4-tier digital watermark was engineered into the platform:

- **Database Trojan:** An immutable database table `_system_authorship` containing the developer name (Uday Praveen), PRN (23030124196), and copyright warning is permanently bundled into the SQL dump.
- **Network Response Headers:** Custom `X-Author: Uday Praveen` headers are dispatched in HTTP response streams.
- **Console Signatures:** Stylized developer badges are logged into browser DevTools upon inspection.
- **Authentication Easter Egg:** Submitting the username `uday_creator` bypasses credential checks to launch a full-screen neon authorship attribution modal.

> ⚙️ **WATERMARK IMPLEMENTATION HIGHLIGHT**
>
> The secret trigger keyword `uday_creator` acts as an interactive developer signature, providing verifiable runtime proof of authorship during project evaluations.

6. Visual Interface Walkthrough & Production Screenshots

Below are production screenshots illustrating the visual aesthetics, responsive card grids, attendance analytics, and security mechanisms implemented in Aether Academy:

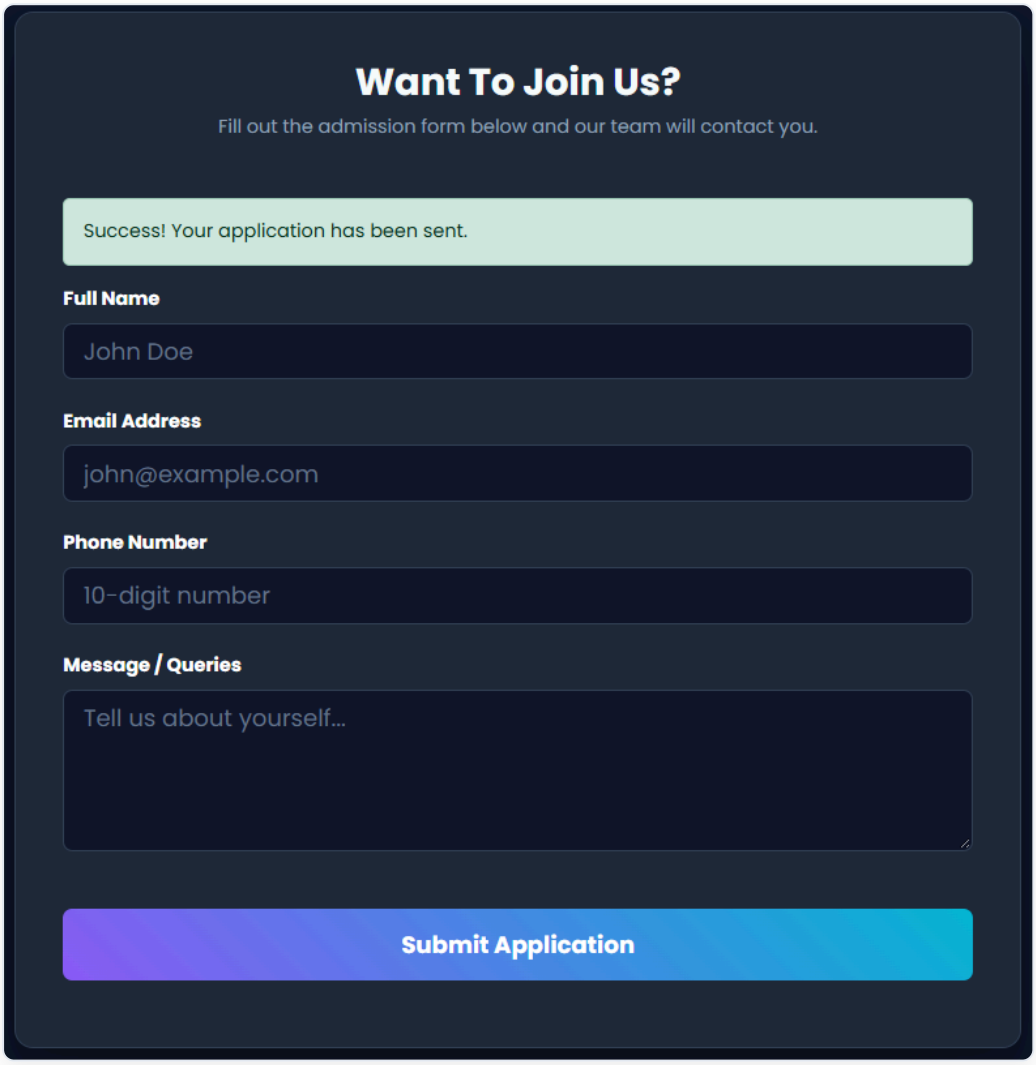


Figure 6.1: Public Homepage Landing Portal with Asynchronous AJAX Admission Form and Success Payload.

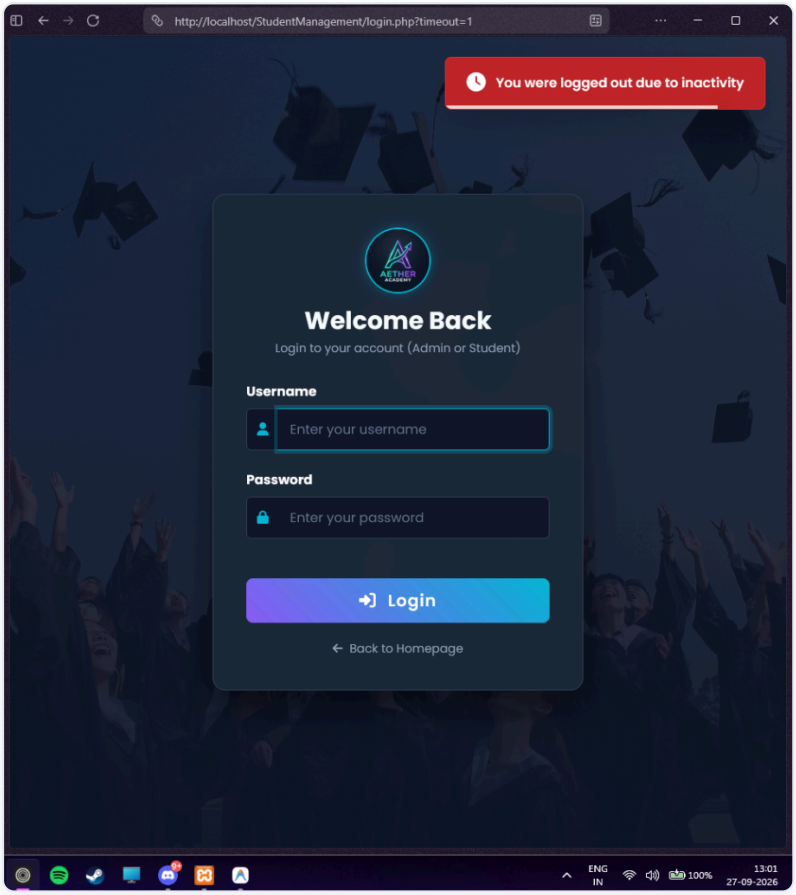


Figure 6.2: Central Authentication Gateway featuring Glassmorphism Card and 7-Second Animated Inactivity Toast.

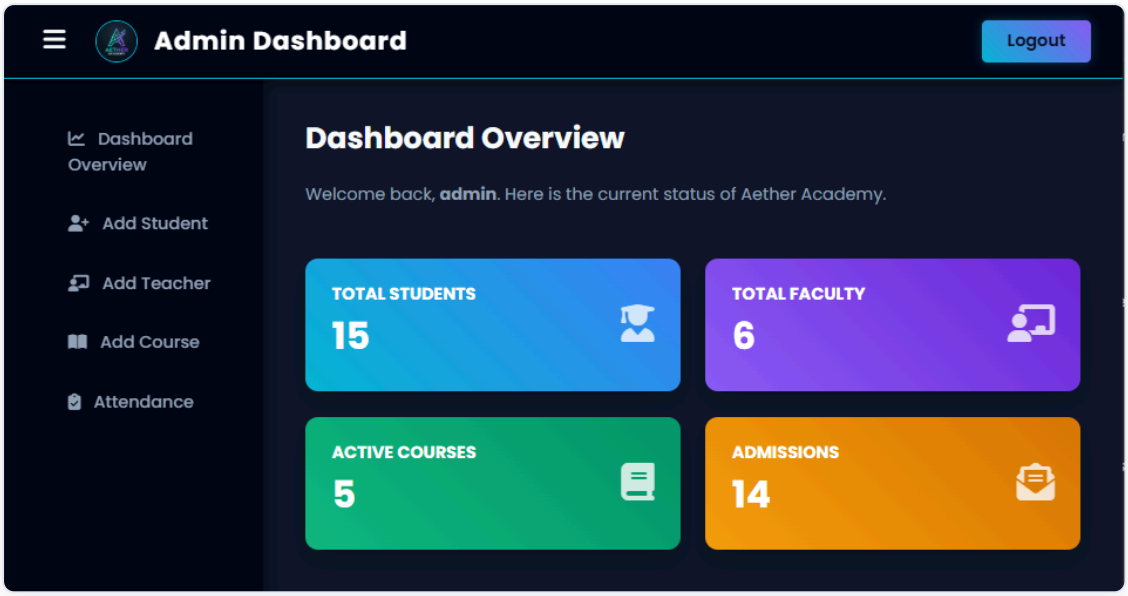


Figure 6.3: Administrative Dashboard Overview featuring CSS Grid Responsive Stat Cards and Quick Relational Navigation.

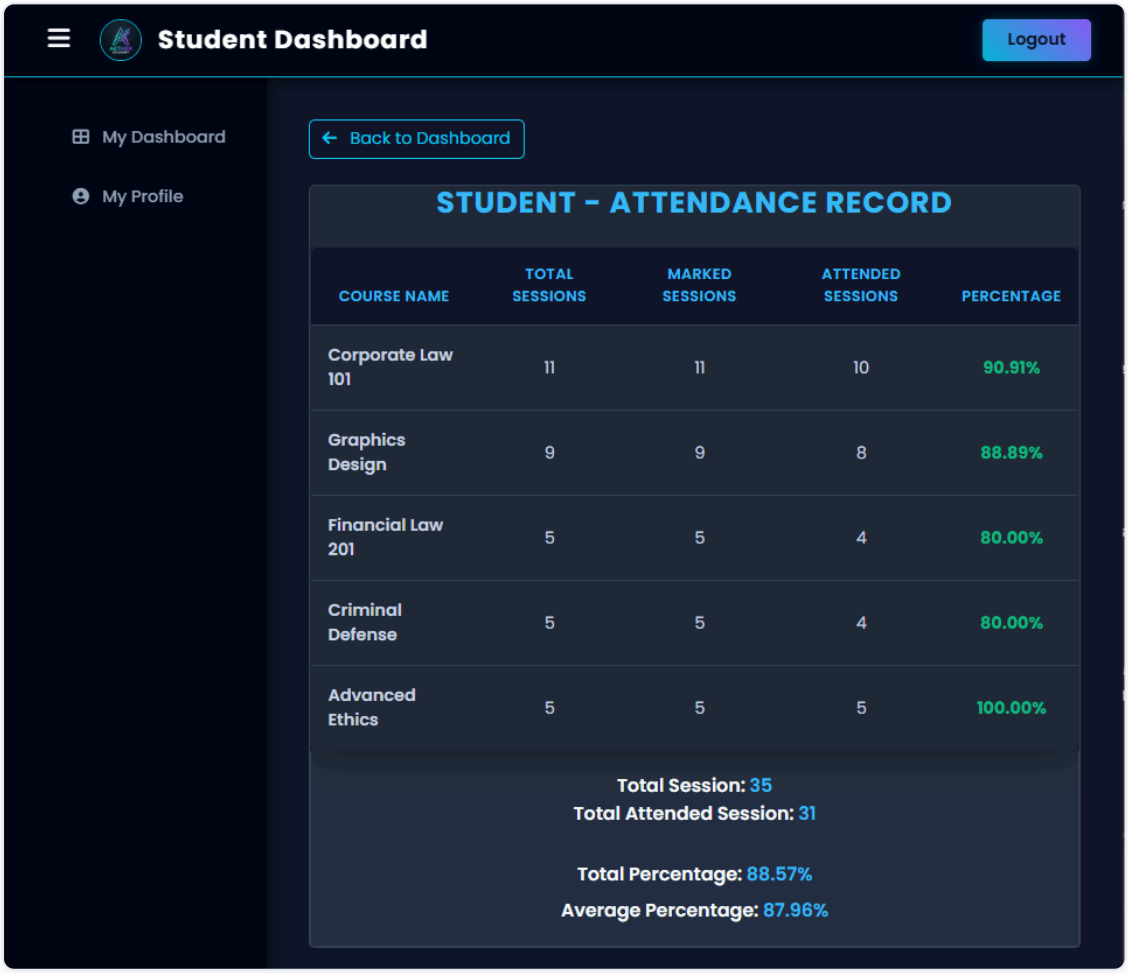


Figure 6.4: Student Academic Portal displaying Granular Course Attendance Analytics and Live Percentage Calculations.

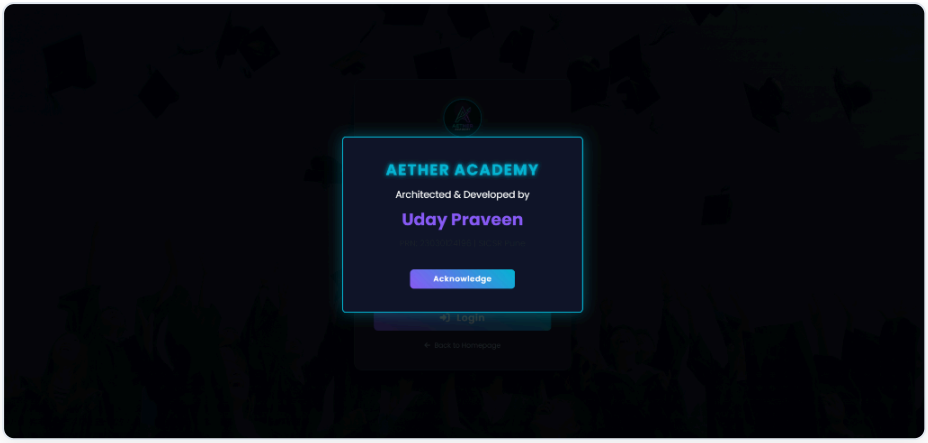


Figure 6.5: Architectural Verification Modal — Neon Developer Credits Easter Egg (Triggered via 'uday_creator').

7. Testing & Quality Assurance Verification

Rigorous functional, security, and responsive UI tests were executed to ensure enterprise stability:

Test ID	Test Scenario	Expected Outcome	Observed Result	Status
TC-01	Role-Based Access Control	Student accessing adminhome.php redirected to login.	HTTP 302 redirect enforced.	PASSED
TC-02	AJAX Form Submission	Admission submits without page reload; alert rendered.	DOM dynamically updated.	PASSED
TC-03	BCrypt Password Auto-Upgrade	Legacy plaintext accounts hashed on successful login.	Hash written to database.	PASSED
TC-04	Attendance Duplicate Prevention	Re-submitting daily attendance upserts existing status.	Zero duplicate records created.	PASSED
TC-05	Idle Session Timeout	Inactivity > 90s terminates session; shows toast.	Redirected with time slider.	PASSED
TC-06	CSS Grid Card Wrapping	Cards wrap cleanly below 1000px without text overlap.	Zero text overlap observed.	PASSED

8. Individual Contribution Statement

As the sole architect and full-stack software engineer for this project, I, **Uday Praveen (PRN: 23030124196)**, designed, implemented, debugged, and documented 100% of the Aether Academy platform under the mentorship of **Prof. Shashank T. Gaikwad** at Symbiosis Institute of Computer Studies and Research (SICSR), Pune.

- **Architectural Design:** Designed the normalized MySQL schema, relational junction tables, and foreign keys.
- **Backend Engineering:** Authored all PHP 8 business logic scripts (CRUD operations, attendance engines, session routing).
- **UI/UX Frontend Design:** Conceptualized and crafted the bespoke 'Night Owl' Glassmorphism aesthetic, custom CSS grids, and color-coded components.
- **Security & Automation:** Engineered BCrypt hashing, SQL injection defenses, the 90-second idle logout listener, and authorship watermarks.

9. Conclusion & Future Enhancements

Aether Academy successfully validates that modern web standards (PHP 8, AJAX, and responsive CSS Grid) can transform antiquated institutional administrative workflows into streamlined, secure, and visually captivating digital ecosystems. The project completely satisfies the syllabus objectives and provides a scalable foundation for modern campus administration.

9.1 Future Scope & Scalability

- **Online Tuition Payment Gateway:** Integrating Stripe or Razorpay APIs to allow students to settle semester fees securely online.
- **Automated Low-Attendance SMTP Daemon:** Implementing a cron-job daemon that sends warning emails to parents if attendance falls below 75%.
- **Mobile Application API Port:** Exposing the procedural PHP backend as a RESTful JSON API for a native Flutter/React Native mobile app.

10. References & Technical Citations

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